Reducing Hospital Costs Through ML-Driven Diabetic Readmission Prevention

Unplanned 30-day hospital readmissions cost the U.S. healthcare system over $26 billion annually, with diabetic patients accounting for a disproportionately large share of avoidable re-hospitalization costs. This project builds a machine learning pipeline to flag high-cost readmission risk at discharge ? giving care coordinators an actionable window to intervene before costs escalate.

Using the UCI Diabetes 130-US Hospitals dataset, I engineered clinically meaningful features, handled severe class imbalance, and optimized for recall over raw accuracy ? because in a cost-containment context, a missed high-risk patient is far more expensive than a false alarm.

Project Structure

```
diabetic-readmission-cost-prevention/
??? diabetic-prediction.md
??? diabetic-prediction.pdf              # Full project documentation
??? Data/
?   ??? diabetic_data.csv
?   ??? IDS_mapping.csv
?   ??? cleaned_data.csv
??? Code.ipynb                           # End-to-end notebook with code & analysis
??? requirements.txt
??? tuned_model.joblib                   # Production-ready saved model
```

Business Problem

Every hospital readmission within 30 days is a signal of discharge risk that wasn't caught in time. For diabetic patients specifically, the complexity of comorbidities, polypharmacy, and post-discharge care gaps makes this prediction especially valuable.

Goal: Identify patients at discharge who are likely to return within 30 days ? so care managers can schedule follow-ups, arrange home health services, or extend observation before costs compound.

Dataset Overview

- Source: UCI Diabetes 130-US Hospitals (1999?2008)
- Records: ~100,000 hospital encounters across 130 U.S. hospitals
- Features: 50 variables covering demographics, diagnoses, procedures, lab results, medications, and discharge details
- Target Variable: Binary ? readmitted within 30 days (<30) vs. not

Methodology

1. Data Cleaning & Standardization

- Removed administrative identifiers with no predictive value
- Replaced "?" placeholders with proper missing value handling
- Filled missing fields by clinical context (e.g., no lab test = "none", not unknown)

2. Cost-Relevant Feature Engineering

- ICD-9 Grouping: Mapped raw diagnosis codes into 12 clinical categories to reduce noise and surface patterns that drive expensive readmissions
- Discharge Risk Flags: Created binary indicators for discharge destination (home vs. facility vs. hospice) ? discharge routing is one of the strongest cost signals
- Insulin & Medication Signals: Encoded treatment intensity as binary features ? under-treated patients at discharge are disproportionately represented in readmissions
- Interaction Features: age ? inpatient history captures the compound risk of older patients with high prior utilization
- Admission Pathway: Flagged emergency admissions, facility transfers, and ER-direct admissions ? these pathways predict both readmission and higher downstream costs

3. Exploratory Data Analysis

- Confirmed 11.1% positive class rate ? severe imbalance requiring explicit handling
- Identified age, prior inpatient visits, and discharge destination as strongest cost-linked signals
- Analyzed medication complexity, lab result severity, and length of stay distributions

4. Model Development

- Algorithms evaluated: Logistic Regression, Random Forest, XGBoost, LightGBM
- Imbalance strategies: class_weight='balanced', scale_pos_weight, SMOTE
- Validation: Stratified 5-fold cross-validation with F1-score optimization
- Interpretability: Feature importance rankings and SHAP values for audit trail

5. Evaluation Framework

Optimized for recall (catching high-risk patients) rather than accuracy, since the cost of missing a readmission far outweighs the cost of a false positive intervention.

Model Performance Summary

Model	Recall	Precision	F1-Score	Accuracy
LightGBM (Tuned)	0.62	0.18	0.28	0.65
XGBoost (Tuned)	0.60	0.18	0.27	0.65
Logistic Regression	0.54	0.17	0.25	0.66
SMOTE + LightGBM (0.1)	0.81	0.16	0.24	0.52

The tuned LightGBM model with class_weight='balanced' offers the best balance of recall and precision for operational deployment.

Top Cost-Driving Risk Factors

Feature	Cost Implication

number_inpatient	Repeat inpatient history is the single strongest p
discharged_to_facility / discharged_home	Facility discharge reduces readmission risk; home
insulin_used	Active insulin adjustments signal unstable glycemi
diabetesMedication	Medication-managed patients need tighter post-disc
age_inpatient_interaction	Elderly patients with high utilization history car
num_lab_procedures	High lab order volume reflects clinical complexity
num_medications	Polypharmacy increases post-discharge adherence ri

Why Precision Is Low ? and Why That's Acceptable

At ~18% precision, the model flags roughly 5?6 patients for every true readmission it catches. In a cost-reduction context, this is defensible:

- A care coordinator follow-up call costs ~$25?$50 in staff time
- An average diabetic readmission costs $14,000?$17,000 in hospital charges
- Even at 18% precision, the expected value per flagged patient is strongly positive

The real limitation is what this dataset can't see: medication adherence after discharge, home support availability, mental health status, and social determinants of health. These gaps explain why precision plateaus, and why this model works best as a triage tool feeding into a broader discharge planning workflow ? not as a standalone predictor.

Installation

```
pip install -r requirements.txt
```

Reproducibility

- Python 3.9+
- All preprocessing encapsulated in sklearn.Pipeline ? no data leakage
- Trained in Jupyter Notebook; model serialized with joblib
- random_state=42 set throughout for reproducibility

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